

A Progressive Polyhedral Approximation Method for Nonlinear PDE-Constrained Electricity-Water Nexus Dispatch

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Abstract—This letter proposes an efficient progressive polyhedral approximation (PA) method to tackle the high nonlinearity and nonconvexity of optimal electricity-water nexus (EWN) dispatch caused by hyperbolic nonlinear partial differential equations (HNPDEs). In this method, the HNPDE-constrained EWN dispatch model can be reformulated into a tractable mixed-integer linear programming (MILP) problem by tailored adaptive discretization and piecewise PA. Furthermore, a progressive approximation refinement technique is developed to dynamically strengthen the MILP for efficient convergence to a near-optimal solution. Comparative studies have validated the effectiveness of the proposed method in reducing decision-making time for the EWN dispatch.

Index Terms—Distribution networks, emergency coordination, PDE-constrained optimization, water-energy nexus.

I. INTRODUCTION

ELECTRICITY-WATER nexus (EWN) is widely recognized as critical lifeline infrastructure to prevent and mitigate flood-induced power outage risks during rainstorms [1]. The emergency dispatch of EWN aims to co-optimize electric drainage pumps in watershed networks (WNs) and dynamic reconfiguration with distributed energy resources in distribution networks (DNs) for system resilience enhancement. The optimal EWN dispatch necessarily involves WN dynamics governed by hyperbolic nonlinear partial differential equations (HNPDEs) to accurately represent the physics of open-channel flows in WNs and to mitigate riverine flood risks to DNs. It should be numerically discretized

and formulated as a mixed-integer nonlinear programming (MINLP) problem for computational solvability. Nevertheless, a discretization of HNPDEs in time and space results in a large number of highly nonlinear, nonconvex algebraic constraints with fractional exponents of water level variables, and hence this MINLP is a challenging optimization problem which cannot be solved by conventional methods. Consequently, this letter strives to develop an efficient solution method to enhance the decision-making efficiency and promote the resilient dispatch of EWN against rainstorms.

Existing solution methodologies for PDE-constrained multi-energy dispatch problems can be categorized as frequency-domain transformation and time-domain discretization [2]. The former transforms the Euler equations of gas and heat pipe flows into frequency-domain energy-circuit equations, where convective and friction terms are neglected or linearized. Differing from pressurized pipe flows in closed conduits, the open-channel flow is a gravity-driven, free-surface, and unsteady liquid flow governed by the Saint-Venant equations (SVEs) [1]. Neglecting the convective term in the SVEs causes inaccuracies in simulating water levels during rainfall, and the open-channel flow velocity lacks a stable nominal value for local linearization. Therefore, the frequency-domain transformation is inapplicable to the EWN dispatch problem. In the latter, PDE-constrained multi-energy dispatch models can be discretized into MINLP problems using finite difference methods (FDMs). Piecewise linearizations and relaxations with uniform partitioning are commonly used to handle nonlinearities, but they incur substantial computational costs [3]. To improve solution efficiency, some studies have developed bound contraction methods for bilinear terms to successively tighten their piecewise relaxations [4]. However, nonconvex constraints with fractional exponents of water level variables, arising from the discretization of the SVEs, make these methods impractical for solving the discrete MINLP formulation of the EWN dispatch model. A comparison between the proposed method and existing related works is summarized in Table I to highlight the advancement of this letter.

In this letter, a progressive polyhedral approximation (PA) method is proposed to efficiently solve the HNPDE-constrained EWN dispatch problem to reduce load shedding and prevent riverine flooding. The main contributions are twofold, as follows: 1) A tailored adaptive FDM is devised to discretize the SVEs of WNs, and then the EWN dispatch model can be reformulated into a tractable MILP problem using a piecewise PA scheme; 2) A progressive approximation refinement technique with dynamic partitioning is developed to iteratively strengthen

Received 21 October 2024; revised 29 January 2025; accepted 12 March 2025. Date of publication 21 March 2025; date of current version 23 April 2025. This work was supported in part by the Innovation and Technology Support Programme of HKSAR Government under Project ITS/044/22MS; in part by the National Natural Science Foundation of China under Grant 52277091; and in part by the Science and Technology Innovation Program of Hunan Province under Grant 2024RC1037. Paper no. PESL-00350-2024. (Corresponding author: Bin Zhou.)

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Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TSG.2025.3553737>.

Digital Object Identifier 10.1109/TSG.2025.3553737

TABLE I
COMPARISONS OF THE PROPOSED METHOD WITH EXISTING WORKS

Ref.	Involved networks	Fluid flow characteristics	PDE constraint handling	Adaptive discretization	Progressive polyhedral approximation
[1]	EN-WN	Open-channel hydrodynamics	TDD without full friction slope terms	-	-
[2]	EN-HN	Closed-pipe thermodynamics	FDT without convective/quadratic terms	-	-
[3]	WDN	Closed-pipe steady state	Modeling without PDE constraints	-	-
[5]	EN-HN	Closed-pipe thermodynamics	TDD without quadratic terms	-	-
[6]	EN-UDN	Closed-pipe hydrodynamics	LT without convective terms	-	-
This letter	EN-WN	Open-channel hydrodynamics	TDD with all nonlinear terms	✓	✓

EN: Electricity network. HN: Heat network. WDN: Water distribution network. UDN: Urban drainage network. FDT: Frequency-domain transformation. TDD: time-domain discretization. LT: Laplace transform. ✓: Item involved. -: Item not involved.

the MILP for efficient convergence to a near-optimal solution, enhancing decision-making efficiency while ensuring solution accuracy.

II. METHODOLOGY

A. HNPDE-Constrained EWN Dispatch Modeling

The topology of a WN can be described as a graph $(\mathcal{J}, \mathcal{R})$, where $\mathcal{J}$ is the set of junctions and $\mathcal{R}$ denotes the set of rivers. The open-channel flow in each river $l \in \mathcal{R}$ can be well-described using the 1-D SVEs, a set of HNPDEs [1]:

$$\partial_t h_l + \partial_x q_l = \delta / w_l, \quad (1a)$$

$$\partial_t q_l + \partial_x \left(\frac{q_l^2}{h_l} + \frac{gh_l^2}{2} \right) = gh_l(S_l^b - S_l^f), \quad (1b)$$

$$S_l^f = \sigma^2 q_l^2 h_l^{-\frac{10}{3}} \left(1 + \frac{2h_l}{w_l} \right)^{\frac{4}{3}}, \quad (1c)$$

where q_l , h_l , and S_l^f are the flow rate, water level, friction slope, respectively, and depend on time $t \in (0, T]$ and space $x \in [0, L_l]$ with T a finite time horizon and L_l the length of the river l ; w_l is the width of river l ; S_l^b is the bottom slope and equals 0 in this letter; δ is the rainfall intensity; σ is the Manning coefficient; and g is the gravitational acceleration.

During rainstorms, DNs strive to minimize load shedding costs and ensure a reliable power supply to pumping station (PS) loads by network reconfiguration with distributed generation, and WNs aims to optimize the operation of PSs to prevent riverine flooding while avoiding operational constraint violation of DNs. Let $\mathcal{T} := \{1, \dots, I\}$ be the set of discrete time periods spanning T , where $I = T/\Delta t$ and Δt is the period duration. Let $\mathcal{L}/\mathcal{B}$ be the set of lines/buses, $\mathcal{K}$ the set of PSs, and $\mathcal{B}_k^{\text{ps}}$ the set of buses that can be connected to PS k . The EWN dispatch problem is modeled as follows:

$$\min \sum_{i \in \mathcal{T}} \sum_{i \in \mathcal{B}} v_i p \Delta P_{i,t} \Delta t + c \sum_{l \in \mathcal{R}} \int_0^{L_l} \int_0^T h_l(x, t) dx dt \quad (2a)$$

$$\text{s.t. (1)} \quad \forall l \in \mathcal{R}, x \in [0, L_l], t \in (0, T] \quad (2b)$$

$$(a_{ij,t} - 1)M \leq V_{i,t} - V_{j,t} - 2(r_{ij}P_{ij,t} + x_{ij}Q_{ij,t}) \\ \leq (1 - a_{ij,t})M \quad \forall (i, j) \in \mathcal{L}, t \in \mathcal{T} \quad (2c)$$

$$P_{k,i,t}^{\text{ps}} \leq e_{k,i,t}^{\text{ps}} \bar{P}_k^{\text{ps}}, \sum_{i \in \mathcal{B}_k^{\text{ps}}} e_{k,i,t}^{\text{ps}} \leq 1 \quad \forall k \in \mathcal{K}, t \in \mathcal{T} \quad (2d)$$

$$(h_l, q_l, S_l^f, \Delta P_{i,t}, a_{ij,t}, P_{ij,t}, Q_{ij,t}, e_{k,i,t}^{\text{ps}}, P_{k,i,t}^{\text{ps}}) \in \Gamma(2e)$$

where $\Delta P_{i,t}$ is the load shedding of bus i at time t ; p is the unit cost of load shedding; v_i is the priority weight of load at bus i ; c is a weighting coefficient; $P_{k,i,t}^{\text{ps}}$ is the active power consumption of PS k supplied from bus i at time t , and $\bar{P}_k^{\text{ps}}$ is its upper limit; and $e_{k,i,t}^{\text{ps}} \in \{0, 1\}$ equals 1 if PS

k is connected to bus i at time t , and 0 otherwise. Variables $V_{i,t}$, $a_{ij,t}$, $P_{ij,t}$, and $Q_{ij,t}$, as well as parameters r_{ij} , x_{ij} , and M are detailed in [1]. Other linear constraints for WNs and DNs have been formulated in [1], defining a feasible region Γ in (2e).

B. MILP Reformulation With Polyhedral Approximation

A tailored adaptive FDM is devised to handle HNPDE constraint (1) for numerically solving the EWN dispatch problem (2). This method employs the Preissmann four-point implicit scheme to discretize the SVEs in space and time, ensuring unconditional computational stability [1]. The continuous space-time domain $[0, L_l] \times [0, T]$ is divided into a mesh of discrete points with time step Δt^m and space step $\Delta x_{n,l}$. Let $\mathcal{T}^+ := \{0\} \cup \mathcal{T}$ be the set of time indices and $\mathcal{N}_l^+ := \{0, 1, \dots, N_l\}$ be the set of space indices for river l . Variables and parameters at point $(n, m) \in \mathcal{N}_l^+ \times \mathcal{T}^+$ are denoted with subscripts ' n ' and ' l ' and a superscript ' m '. Instead of using fixed time and space steps, Δt^m and $\Delta x_{n,l}$ are adaptively adjusted based on the rainfall intensity at time m and the distance from space point n of river l to its nearest critical outdoor DN component. In periods of high rainfall intensity, a small Δt^m (e.g., 5 to 15 minutes) is chosen to enable swift mitigation of riverine flood risks to DNs. When rainfall intensity decreases significantly, a large Δt^m (e.g., 30 to 60 minutes) is used to reduce computational burdens. Finer space steps are assigned to river sections proximal to critical outdoor DN components, where river overflow significantly threatens DN security. In contrast, coarser space steps are used in sections farther from DN components to improve computational efficiency, thereby ensuring practicality for large-scale applications. Let $\mathcal{C}_l$ be the set of DN components near river l , $\Delta x_{n,l}$ is given by $\Delta x_{n,l} = \min\{\Delta \bar{x}_l - (\Delta \bar{x}_l - \Delta \underline{x}_l)e^{-(d_{n,c^*,l}/d_{c^*,l}-1)}, \tilde{L}_l\}$, where $\tilde{L}_l = L_l - \sum_{i=0}^{n-1} \Delta x_{i,l}$, $c^* = \arg \min_{c \in \mathcal{C}_l} \{d_{n,c,l}\}$, $\Delta \bar{x}_l/\Delta \underline{x}_l$ is the upper/lower bound of $\Delta x_{n,l}$, $d_{n,c,l}$ is the distance from point n of river l to DN component c , and $d_{c,l}$ is the distance from DN component c to river l . Each space point n of river l is determined iteratively by adding $\Delta x_{n,l}$ to the position of point $n-1$. The discretization of (1) yields:

$$\frac{h_{n+1,l}^{m+1} + h_{n,l}^{m+1} - h_{n+1,l}^m - h_{n,l}^m}{2\Delta t^m} + \frac{q_{n+1,l}^{m+1} - q_{n,l}^{m+1}}{\Delta x_{n,l}} = \tilde{\delta}_l, \quad (3a)$$

$$\frac{q_{n+1,l}^{m+1} + q_{n,l}^{m+1} - q_{n+1,l}^m - q_{n,l}^m}{2\Delta t^m} + \frac{g(\alpha_{n+1,l}^{m+1} - \alpha_{n,l}^{m+1})}{2\Delta x_{n,l}} \\ = \frac{\psi_{n,l}^{m+1} - \psi_{n+1,l}^{m+1}}{\Delta x_{n,l}} - g\sigma^2(\phi_{n+1,l}^{m+1} + \phi_{n,l}^{m+1}), \quad (3b)$$

$\forall l \in \mathcal{R}, n \in \mathcal{N}_l^+ \setminus \{N_l\}, m \in \mathcal{T}^+ \setminus \{I\}$. Here, $\tilde{\delta}_l := \frac{\delta^{m+1}}{2w_{n+1,l}} + \frac{\delta^{m+1}}{2w_{n,l}}$, and variables $\alpha_{n,l}^m$, $\psi_{n,l}^m$, and $\phi_{n,l}^m$ are defined as:

$$\alpha_{n,l}^m = (h_{n,l}^m)^2, \quad \phi_{n,l}^m = \omega_{n,l}^m \cdot \beta_{n,l}^m, \quad \psi_{n,l}^m = \zeta_{n,l}^m \cdot \beta_{n,l}^m, \quad (4)$$

where

$$\zeta_{n,l}^m = \frac{1}{h_{n,l}^m}, \quad \beta_{n,l}^m = (q_{n,l}^m)^2, \quad \omega_{n,l}^m = 0.5(h_{n,l}^m)^{-\frac{7}{3}} \left(1 + \frac{2h_{n,l}^m}{w_{n,l}} \right)^{\frac{4}{3}}. \quad (5)$$

Algorithm 1: Progressive Approximation Refinement

Input: $LB = 0$, $UB = 10^9$; Global optimality tolerance ε ; Tuning parameters τ , κ , μ ; $\mathcal{P}_x := \{(\bar{x}, \bar{x})\}$, $\forall x \in X$

Output: A near-optimal solution to the discrete formulation of (2)

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1 while  $(UB - LB)/UB \times 100\% > \varepsilon$  do
2   Solve the MILP (6) with  $\{\mathcal{P}_x\}_{x \in X}$  and derive  $\{x^*\}_{x \in X}$ ,  $\{y^*\}_{y \in Y}$ ,
   and its objective value  $obj^*$ ;  $LB \leftarrow obj^*$ ;
3   foreach  $x \in X$  do
4     Identify the active partition  $(l_x, u_x)$  in  $\mathcal{P}_x$ ;
5     if  $\min\{(x^* - l_x), (u_x - x^*)\} > \mu$  then
6        $a \leftarrow l_x + (x^* - l_x)/\tau$ ,  $b \leftarrow u_x - (u_x - x^*)/\tau$ ;
7        $\mathcal{P}_x \leftarrow (\mathcal{P}_x \setminus \{(l_x, u_x)\}) \cup \{(l_x, a), (a, x^*), (x^*, b), (b, u_x)\}$ ;
8     else
9       Identify the largest partition  $(\hat{l}, \hat{u})$  in  $\mathcal{P}_x$ ;
10       $\mathcal{P}_x \leftarrow (\mathcal{P}_x \setminus \{(\hat{l}, \hat{u})\}) \cup \{(\hat{l}, \hat{l} + (\hat{u} - \hat{l})/2), (\hat{l} + (\hat{u} - \hat{l})/2, \hat{u})\}$ ;
11    end
12  end
13  Solve NLP problem (7) to local optimality and derive its objective
  value  $obj$ ; Save the obtained solution as the warm start for the
  MILP (6) at next iteration;
14   $UB \leftarrow \min\{obj, UB\}$ ;  $\tau \leftarrow \tau - \kappa$ ;
  
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min (6a)
s.t. (2c) – (5); $l_x \leq x \leq u_x, \forall x \in X; y = y^*, \forall y \in Y$. (7)

Thus, HNPDE constraint (1) is transformed into the linear algebraic one (3) and the nonlinear nonconvex algebraic ones (4) and (5), leading to a discrete MINLP formulation of (2). $\alpha_{n,l}^m$, $\omega_{n,l}^m$, and $\zeta_{n,l}^m$ are nonlinear convex functions of $h_{n,l}^m$ over $0 < h_l \leq h_{n,l}^m$. $\beta_{n,l}^m$ is a nonlinear convex function of $q_{n,l}^m$, and $\omega_{n,l}^m \cdot \beta_{n,l}^m$ and $\zeta_{n,l}^m \cdot \beta_{n,l}^m$ are bilinear terms. A piecewise PA scheme is employed to address (4) and (5). In this scheme, the domains of $h_{n,l}^m$, $q_{n,l}^m$, and $\beta_{n,l}^m$ are divided into a set of partitions, and piecewise-linear envelopes are applied to nonlinear convex functions [3], while piecewise McCormick envelopes are applied to bilinear terms [5]. These envelopes collectively form a piecewise PA that tightly encloses the feasible region defined by (4) and (5). As a result, the MILP reformulation of (2) is given by

$$\min \sum_{m \in \mathcal{T}} \sum_{i \in \mathcal{B}} v_i p \Delta P_{i,m} \Delta t^m + c \sum_{l \in \mathcal{R}} \sum_{n \in \mathcal{N}_l^+} \sum_{m \in \mathcal{T}} h_{n,l}^m \quad (6a)$$

$$\text{s.t. (2c)–(3), } \widehat{\beta}_{n,l}^m(q_{n,l}^m), \widehat{\alpha}_{n,l}^m(h_{n,l}^m), \widehat{\omega}_{n,l}^m(h_{n,l}^m) \quad (6b)$$

$$\widehat{\zeta}_{n,l}^m(h_{n,l}^m), \widehat{\phi}_{n,l}^m(\omega_{n,l}^m, \beta_{n,l}^m), \widehat{\psi}_{n,l}^m(\zeta_{n,l}^m, \beta_{n,l}^m) \quad (6c)$$

where $\widehat{y}(x)$ denotes the piecewise linear envelopes for the nonlinear convex function $y = f(x)$, and $\widehat{z}(x, y)$ denotes the piecewise McCormick envelopes for the bilinear term $z = xy$.

C. Progressive Approximation Refinement Technique

Refining the piecewise PAs is essential for strengthening the MILP (6) and enhancing solution accuracy by increasing partitions in the domains of $h_{n,l}^m$, $q_{n,l}^m$, and $\beta_{n,l}^m$. However, while numerous uniform partitions improve solution accuracy, they significantly increase auxiliary binary variables, imposing heavy computational burdens when solving the MILP (6). To address this issue, a progressive approximation refinement technique with dynamic partitioning that focuses on the active partitions of $h_{n,l}^m$, $q_{n,l}^m$, and $\beta_{n,l}^m$ is developed to iteratively strengthen the MILP (6), as detailed in Algorithm 1. Here,

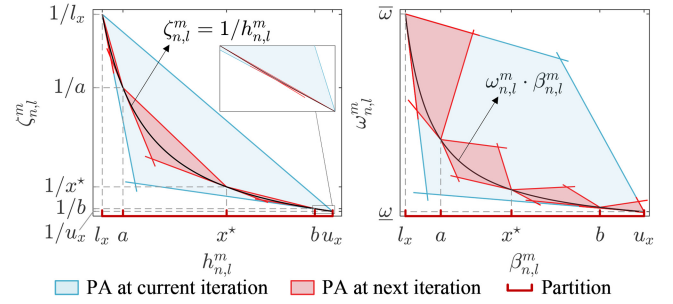


Fig. 1. PA refinement for the active partition of $\zeta_{n,l}^m = 1/h_{n,l}^m$ and $\omega_{n,l}^m \cdot \beta_{n,l}^m$.

TABLE II
COMPARATIVE PERFORMANCE RESULTS OF SCHEMES 1–7

Scheme	Load shedding (MWh)	ROG (%)	Solution time (s)	MSD (%)
1	4.760	0.010	524	0.022
2	4.797	0.091	2759	0.041
3	4.843	0.093	2840	0.055
4	4.155	9.602	63	10.3
5	4.778	0.006	837	0.479
6*	N/A	N/A	>9999	N/A
7	4.806	0.008	2881	1.515
Baseline	4.761	-	1509	-

* The SCIP solver in Scheme 6 fails to converge to a solution.

$X := \{h_{n,l}^m, q_{n,l}^m, \beta_{n,l}^m\}_{n \in \mathcal{N}_l^+, m \in \mathcal{T}^+, l \in \mathcal{R}}$, Y is the set of binary variables in (2), $\mathcal{P}_x$ is the set of partitions for $x \in X$, and x^*/y^* denotes the optimal solution for $x \in X/y \in Y$. Initially, $\mathcal{P}_x$ for each $x \in X$ consists of a single partition $[\bar{x}, \bar{x}]$, where $\bar{x}$ and $\bar{x}$ are the lower and upper bounds of x , efficiently obtained by parallel optimality-based bound tightening [3]. In line 2, the MILP (6) is solved, with its optimal objective value obj^* serving as a valid lower bound LB for the MINLP's objective value, as the MINLP's feasible region is a subset of the MILP (6)'s. In line 4, the active partition in $\mathcal{P}_x$ for each $x \in X$, denoted as (l_x, u_x) , is identified as the partition containing x^* . Lines 5–9 split the active partition and the largest partition to refine the piecewise PAs. If (l_x, u_x) is sufficiently large (i.e., the condition in line 5 is satisfied), it is split into four nonuniform partitions. Otherwise, the largest partition in $\mathcal{P}_x$, denoted as $(\hat{l}, \hat{u})$, is split into two equal partitions. This ensures that no regions in the domain of x remain unexplored, maintaining the exhaustiveness of the progressive approximation refinement [7]. Fig. 1 illustrates the PA refinement for the active partition using $\zeta_{n,l}^m = 1/h_{n,l}^m$ and $\phi_{n,l}^m = \omega_{n,l}^m \cdot \beta_{n,l}^m$ as examples. In line 12, the NLP problem (7) is solved efficiently to obtain a locally optimal solution using solvers such as IPOPT [5]. As the feasible region of the NLP is a strict subset of the MINLP's, obj serves as a valid upper bound UB for the MINLP's objective value. As the iterations progress, the piecewise PAs are progressively refined, leading to an increasingly tighter MILP (6) of the MINLP. Consequently, obj^* monotonically increases and converges towards the optimal objective value of the MINLP. The iterations terminate when the optimality gap $(UB - LB)/UB \times 100\%$ is less than or equal to a predefined tolerance ε , yielding an ε -global optimum as a near-optimal solution to the discrete MINLP formulation of (2).

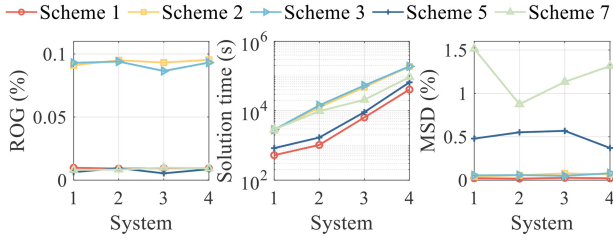


Fig. 2. Performance comparison of Schemes 1-3, 5, and 7 in different scale systems: System 1 (33-bus DN + 13-junction WN), System 2 (118-bus DN + 31-junction WN), System 3 (300-bus DN + 96-junction WN), and System 4 (1200-bus DN + 253-junction WN).

III. CASE STUDIES

In this letter, a 13-junction WN in Hunan Province, China, and a modified IEEE 33-bus DN are presented to validate the performance of the proposed method, as referenced in [8]. ε , τ , κ , and μ are set to 0.01%, 5, 0.05, and 0.001, respectively. The EWN dispatch is carried out over $T = 5$ hours. Seven schemes are considered for comparative analysis: Scheme 1 is the proposed method; Scheme 2 solves the MINLP using the sequential solution method from [5]; Scheme 3 addresses (4) and (5) using the multi-dimensional piecewise linearization method from [11]; Scheme 4 converts HNPDE constraint (1) into linear algebraic constraints using the Laplace transform-based PDE simplification method from [6]; and Schemes 5–7 employ various solvers to solve the MINLP, including SHOT + Gurobi [9], SCIP, and XPRESS. A baseline scheme is introduced to provide a benchmark for evaluating the maximum solution differences (MSDs) of $h_{n,l}^m$ and the relative optimality gaps (ROGs) of Schemes 1–7, which solves the MINLP to global optimality using Gurobi. All schemes are tested in the PyCharm environment with Python 3.12.4 and GAMS Py 1.5.0 on a computer equipped with an Intel Core i9-14900KS CPU and 192GB of DDR5 RAM. Table II summarizes the comparative performance statistics for Schemes 1–7. Unlike Schemes 2 and 3, which require a large number of uniform partitions, the proposed method significantly reduces the number of partitions, reducing solution time by 81.01% and 81.55%, with an acceptable MSD of 0.022%. Although Scheme 4 has the shortest solution time, it causes significant water level deviation. Using this scheme in actual dispatch could lead to the underestimation of water levels, increasing riverine flood risks to DNs. Compared to Schemes 5–7, the proposed method achieves both shorter solution time and smaller MSD. Fig. 2 shows the performance of Schemes 1–3, 5, and 7 as the system

scale of EWN increases. Scheme 1 achieves the shortest solution time and highest accuracy compared to the other schemes, highlighting the effectiveness and superiority of the proposed method for large-scale EWN dispatch applications.

IV. CONCLUSION

This letter proposes an efficient progressive PA method to address the high nonlinearity and nonconvexity of the optimal EWN dispatch problem caused by HNPDEs. Compared to conventional multi-dimensional piecewise linearization methods, the proposed method can reduce solution time by 81.55% with a solution difference of less than 0.022%. The case study results demonstrate the effectiveness and superiority of the proposed method in enhancing the decision-making efficiency and solution accuracy of the EWN dispatch. The proposed method potentially encounter computational challenges when applied to extra large-scale EWN dispatch. Future work will focus on using reduced-order SVEs to reduce solution complexity and improve scalability in extra large-scale applications.

REFERENCES

- [1] Y. Cao, B. Zhou, C. Y. Chung, T. Wu, L. Zheng, and Z. Shuai, "A coordinated emergency response scheme for electricity and watershed networks considering spatio-temporal heterogeneity and volatility of rain-storm disasters," *IEEE Trans. Smart Grid*, vol. 15, no. 4, pp. 3528–3541, Jul. 2024.
- [2] B. Chen et al., "Energy-circuit-based integrated energy management system: Theory, implementation, and application," *Proc. IEEE*, vol. 110, no. 12, pp. 1897–1926, Dec. 2022.
- [3] B. Tasseff et al., "Polyhedral relaxations for optimal pump scheduling of potable water distribution networks," *INFORMS J. Comput.*, vol. 36, no. 4, pp. 1040–1063, Jul.–Aug. 2024.
- [4] J. Cao, B. Yang, C. Y. Chung, Y. Gong, and X. Guan, "Distributionally robust management of hybrid energy station under exogenous-endogenous uncertainties and bounded rationality," *IEEE Trans. Sustain. Energy*, vol. 15, no. 2, pp. 884–902, Apr. 2024.
- [5] X. Tian, X. Lin, W. Zhong, Y. Zhou, and F. Cong, "Optimal dispatch of integrated electricity and heating systems considering the quality-quantity regulation of heating systems to promote renewable energy consumption," *Energy*, vol. 300, Aug. 2024, Art. no. 131599.
- [6] Y. Cao, B. Zhou, C. Y. Chung, K. Zhou, L. Zhu, and Z. Shuai, "Resilience-oriented coordinated topology reconfiguration of electricity and drainage networks with distributed mobile emergency resources," *IEEE Trans. Smart Grid*, vol. 16, no. 1, pp. 786–800, Jan. 2025.
- [7] R. Horst and H. Tuy, *Global Optimization: Deterministic Approaches*. Berlin, Germany: Springer, 2013.
- [8] Oct. 2024, "Details for case studies." Dataset, Figshare. doi: 10.6084/m9.figshare.27263259.v3.
- [9] A. Lundell and J. Kronqvist, "Polyhedral approximation strategies for nonconvex mixed-integer nonlinear programming in SHOT," *J. Glob. Optim.*, vol. 82, no. 4, pp. 863–896, Mar. 2022.